

TEAM PROGRESS REPORT

AI-Based Network Attack Forecasting from Network Traffic Data

Day 1 Progress Report and Day 2 Execution Plan

CICIDS2017 | DDoS early-warning forecasting pipeline | Team of 6

REPORT DATE	1 September 2026
DATASET	CICIDS2017
GIT BRANCH	feature/pulkit-integration
LATEST COMMIT	ec43af8
COMMIT MESSAGE	Add ML forecasting pipeline and SIH dashboard
REPOSITORY	github.com/pulkitokte/SIH26153-Network-Attack-Forecasting
PURPOSE	Preserve Day 1 results and brief the team for Day 2

THIS DOCUMENT DISTINGUISHES THE ORIGINAL ROADMAP FROM ACTUAL DAY 1 WORK.

Do not treat planned responsibilities as completed ownership. Multi-horizon GRU is not claimed as trained.

1. Purpose of this document

This PDF is an internal SIH 2026 backup deliverable. It preserves the team's complete Day 1 state as of 1 September 2026 and converts that state into an adapted Day 2 execution plan. A teammate who reads only this file should understand who the six members are, what was originally planned, what was actually built, which model numbers are real, what remains incomplete, and what each person must do tomorrow.

How to read this report

ORIGINAL DAY 1 ROADMAP describes planned kickoff responsibilities. ACTUAL DAY 1 PROGRESS describes verified completed work. Those two sections are not interchangeable. Individual names in the original roadmap are role assignments, not automatic ownership of every later ML result.

Repository: <https://github.com/pulkitokte/SIH26153-Network-Attack-Forecasting>

Current branch: feature/pulkit-integration **Latest pushed commit:** ec43af8

The latest work is on the feature branch. This document does not claim that **main** already contains it.

Contents

1. Purpose of this document
2. Team and role mapping
3. Original Day 1 roadmap
4. Actual Day 1 progress
5. Model comparison
6. Website and GitHub
7. Day 1 completion status
8. Day 2 adapted execution plan
9. Day 2 critical path, timeline, and definition of done
10. Risks and honest limitations

2. Team and role mapping

Authoritative mapping from the team roadmap. Primary is the default Day 2 ownership. Secondary is support / overlap. Pragati is a team member. CICIDS2017 is the dataset.

Member	Primary role	Secondary role
Pranshu	Cybersecurity and Network Lead	MITRE mapping and attack scenarios
Pulkit	Web / Dashboard and System Integration Lead	Backend / API and final integration
Pragati	Data Engineering / Data Analyst	Dataset and preprocessing
Ankita	ML / Data Science Lead	World Model experimentation
Riddhi	ML / Data Science and Evaluation	Explainability and benchmarking
Priyanshi	Python Engineering	ML support, automation, integration

6

Team members

CICIDS2017

Dataset

DDoS

Day 1 focus

ec43af8

Latest commit

3. ORIGINAL DAY 1 ROADMAP

These were the original kickoff tasks. They remain useful for role clarity, but they are not a complete description of what was actually finished today. Actual completed work is in Section 4.

Member	Original Day 1 plan	Original Day 2 plan
Pranshu	Problem-statement deep dive; MITRE ATT&CK research; DDoS, port scan, brute force; demo attack scenario concept	Document 3-4 concrete attack patterns and traffic signatures
Pulkit	GitHub repository setup; Flask/FastAPI boilerplate; dashboard boilerplate; draft API contract	Finalise /predict, /traffic-stream, /alerts, /model-stats; start dashboard layout with mock data
Pragati	Download/explore CICIDS2017 / NSL-KDD; document feature list	Feature analysis: flow duration, byte rate, protocol type, flags; identify key features
Ankita	Research Isolation Forest and LSTM; decide TensorFlow vs PyTorch	Rough Isolation Forest baseline
Riddhi	Precision, Recall, F1, ROC-AUC; SHAP/LIME research	Evaluation-script skeleton
Priyanshi	Shared Python environment; requirements.txt / venv; initial data-loading support	Data-loading automation; time-based train/test split automation

Roadmap vs reality

Day 1 work went well beyond this kickoff list. A leakage-safe forecasting pipeline, GRU models, early-warning analysis, multi-horizon windows, and a React dashboard now exist. Day 2 must be replanned around that actual state, not around Isolation Forest as if no sequence model existed.

4. ACTUAL DAY 1 PROGRESS

This section records verified Day 1 outcomes. It is intentionally separate from the original roadmap. Where a result is not confirmed, this report says so instead of inventing a number.

4.A Dataset and data pipeline

The working direction is CICIDS2017 network traffic with a DDoS-focused forecasting and early-warning objective. Completed pipeline work included dataset auditing, exploration, label inventory, DDoS-focused inspection, DDoS feature preparation, temporal feature analysis, forecasting-window construction, leakage / observation safety validation, sequence tensor preparation, and validation reports.

Local workspace folder	Role
raw/	Source CICIDS2017 captures (not pushed)
processed/	Prepared features, windows, predictions (large files excluded)
scripts/	Audit, windowing, training, and evaluation scripts
reports/	Text validation and training reports
models/	Trained model files (*.pt excluded from Git)
samples/	Small inspection samples
website/	React + Vite dashboard copied into the repository

Large raw datasets and generated binary artifacts were intentionally not pushed to GitHub. That is a storage/safety choice, not missing work.

4.B Forecasting Window V1

The first forecasting windows asked whether DDoS appears in the next 100 flows. This version was audited and then retired as insufficient for genuine early warning because observation windows could already contain attack-state information.

Item	Value
Total windows	399

Item	Value
Negative (no DDoS in next 100)	189
Positive (DDoS in next 100)	210
Leakage / overlap checks	Performed
Decision	Insufficient for genuine early-warning; move to V2

4.C Genuine early-warning Window V2

V2 enforces a stricter contract: the 100-flow observation window must contain zero DDoS, and every positive target must correspond to future DDoS. Episode identity is the split unit, not a random row shuffle.

Item	Value
Total windows	1,911
Episodes	21
Windows per episode	91 (81 negative, 10 positive)
Overall negatives / positives	1,701 / 210
Observation windows containing DDoS	0
Every positive window contains future DDoS	Yes
TRAIN / VALIDATION / TEST windows	1,365 / 273 / 273
TRAIN episodes	1-15
VALIDATION episodes	16-18
TEST episodes	19-21
Episode overlap	None
TRAIN targets 0 / 1	1,215 / 150
VALIDATION targets 0 / 1	243 / 30
TEST targets 0 / 1	243 / 30

4.D Sequence tensors (V2)

Tensor	Shape	Checks
X_train	(1365, 100, 68)	NaN = 0, Inf = 0
y_train	(1365,)	Aligned to TRAIN episodes 1-15
X_validation	(273, 100, 68)	Sequence length = 100
y_validation	(273,)	Feature count = 68
X_test	(273, 100, 68)	No episode overlap
y_test	(273,)	Metadata order matches tensors

4.E Logistic Regression baseline

A classical baseline was trained on aggregated temporal features (136 features). The validation threshold was 0.90 (validation F1 = 0.5672). This is the non-sequence reference point, not the operational early-warning model.

Metric	TEST value
Confusion matrix [[TN, FP], [FN, TP]]	[[203, 40], [12, 18]]
Accuracy	0.81
DDoS Precision	0.31

Metric	TEST value
DDoS Recall	0.60
DDoS F1	0.4091
ROC-AUC	0.8514
PR-AUC	0.3014
False Alarm Rate	0.1646

4.F GRU V1

GRU V1 is a sequence model over 100 historical flows x 68 features, predicting DDoS in the next 100 flows. Class weight used training counts only (1,215 negatives / 150 positives; pos_weight = 8.1).

Item	Value
Architecture	GRU(68, hidden=64, layers=2, dropout=0.25); classifier 64 to 32 to 1
Best validation loss	0.2359 (early stopping)
Selected validation threshold	0.65
Validation F1	0.8955
TEST confusion matrix	[[213, 30], [0, 30]]
TEST Accuracy / Precision / Recall / F1	0.89 / 0.50 / 1.00 / 0.6667
TEST ROC-AUC / PR-AUC / FAR	0.9444 / 0.5370 / 0.1235
Model file	models/ddos_gru_forecaster.pt

4.G GRU V1 early-warning (TEST episodes 19-21)

Episode	Attack start	First warning	Lead (flows)	Probability
19	36910	36709	201	0.9517
20	38910	38799	111	0.9681
21	40910	40909	1	0.8843

Summary metric	Value
Episodes tested / warned	3 / 3
Early-warning rate	100% of episodes received some warning
Mean / median lead	104.33 / 111 flows
Min / max lead	1 / 201 flows
>= 200 flows early	1/3 = 33.33%
>= 100 flows early	2/3 = 66.67%
>= 50 flows early	2/3 = 66.67%
>= 10 flows early	2/3 = 66.67%
>= 1 flow early	3/3 = 100%

4.H GRU V2

Item	Value
Architecture	GRU(68, hidden=96, layers=2, dropout=0.30); LayerNorm(96); 96 to 48 to 1 + ReLU + Dropout 0.30
Training budget	Up to 60 epochs; early stopping

Item	Value
Best validation loss	0.2337
Selected validation threshold	0.40
Validation F1	0.8923
TEST confusion matrix	[[215, 28], [0, 30]]
TEST Accuracy / Precision / Recall / F1	0.90 / 0.5172 / 1.00 / 0.6818
TEST ROC-AUC / PR-AUC / FAR	0.9329 / 0.5420 / 0.1152
Model file	models/ddos_gru_forecaster_v2.pt
Predictions	processed/gru_v2_test_predictions.csv

4.I GRU V2 early-warning (TEST episodes 19-21)

Episode	Attack start	First warning	Lead (flows)	Probability
19	36910	36719	191	0.9642
20	38910	38799	111	0.8764
21	40910	40909	1	0.4160

Summary metric	Value
Episodes tested / warned	3 / 3
Early-warning rate	100% of episodes received some warning
Mean / median lead	101 / 111 flows
Min / max lead	1 / 191 flows
>= 200 flows early	0/3 = 0%
>= 100 flows early	2/3 = 66.67%
>= 50 / >= 10 flows early	2/3 = 66.67%
>= 1 flow early	3/3 = 100%

Critical interpretation

Do not present a 100% warning rate as reliable or consistently useful early warning. One held-out episode only received a 1-flow warning. False alarms remain material (GRU V2 TEST FAR = 0.1152). Classification recall of 1.00 is not the same as operational lead time.

4.J Multi-horizon forecasting windows

Multi-horizon windows forecast DDoS presence in the next 50, 100, 200, and 500 flows after a 100-flow benign observation. Dataset generation is complete. Multi-horizon GRU training and evaluation are not claimed as complete in this Day 1 backup.

Item	Value
Horizons	50 / 100 / 200 / 500 flows
Windows per episode / episodes / total	901 / 21 / 18,921
Observation windows containing DDoS	0
Next 50: no DDoS / DDoS	17,871 / 1,050
Next 100: no DDoS / DDoS	16,821 / 2,100
Next 200: no DDoS / DDoS	14,721 / 4,200
Next 500: no DDoS / DDoS	8,421 / 10,500

Item	Value
TRAIN / VALIDATION / TEST	13,515 / 2,703 / 2,703
Episode split	TRAIN 1-15; VALIDATION 16-18; TEST 19-21
Output	processed/ddos_multihorizon_windows_v1.csv
Day 2 implication Day 2 should concentrate on multi-horizon validation, sequence tensors, multi-horizon GRU training, evaluation at 50/100/200/500-flow horizons, and comparison with Logistic Regression, GRU V1, and GRU V2. No multi-horizon GRU metrics are invented here.	

5. Website and GitHub integration

5.A Website (frontend / dashboard)

A React + TypeScript + Vite cybersecurity dashboard was created in Cursor at C:\Users\Pulkit Okte\OneDrive\Desktop\SIH-26153 and copied into the Git repository under website/. Stack: React, TypeScript, Vite, React Router, Recharts, Lucide React, Tailwind CSS.

Pages	Selected components / systems
Dashboard, Alerts, Analytics, Attack Forecast, Live Traffic, Model Performance, Settings, System Architecture, Threat Intelligence	AlertCard, ArchitectureFlow, AnomalyChart, ProtocolChart, ForecastChart, PredictionCard, KPI cards, model metric cards, pipeline diagram, simulation controls, traffic chart, Sidebar, Topbar, RiskBadge, DataTable, SimulationContext, API service, mock data

Dashboard boundary

The dashboard currently represents frontend/dashboard work. This report does not claim that it is connected to a real-time ML backend. Mock or static values must not be presented as live model output.

5.B GitHub integration

Item	Value
Repository	SIH26153-Network-Attack-Forecasting
URL	https://github.com/pulkitokte/SIH26153-Network-Attack-Forecasting
Branch with new work	feature/pulkit-integration
Latest commit	ec43af8
Commit message	Add ML forecasting pipeline and SIH dashboard
Diff	104 files changed; 26,736 insertions; 210 deletions
Push	988185f..ec43af8 to origin/feature/pulkit-integration
main branch	Not claimed to contain this work

.gitignore intentionally excludes node_modules/, dist/, environment files, Python caches, raw CICIDS2017 data, *.npy, *.pt, *.joblib, local processed datasets, and temporary files.

A temporary accidental file ml/scripts/Untitled contained only the text validate_multihorizon_windows.py and was deleted before the commit.

6. Model comparison (confirmed Day 1 TEST results)

Window-level forecasting classification on the held-out TEST split. Multi-horizon GRU is omitted because it is not a confirmed Day 1 trained model in this backup.

Metric	LogReg baseline	GRU V1	GRU V2
Input	136 aggregated features	(100, 68) sequence	(100, 68) sequence
Val. threshold	0.90	0.65	0.40
Val. F1	0.5672	0.8955	0.8923
TEST Accuracy	0.81	0.89	0.90
TEST Precision (DDoS)	0.31	0.50	0.5172
TEST Recall (DDoS)	0.60	1.00	1.00
TEST F1 (DDoS)	0.4091	0.6667	0.6818
TEST ROC-AUC	0.8514	0.9444	0.9329
TEST PR-AUC	0.3014	0.5370	0.5420
TEST FAR	0.1646	0.1235	0.1152
TEST TP / FN	18 / 12	30 / 0	30 / 0
TEST FP	40	30	28

Episode-level early warning is a different question from F1. GRU V1 vs GRU V2 lead times:

Temporal metric	GRU V1	GRU V2
Episodes warned	3/3	3/3
Mean lead (flows)	104.33	101
Median lead (flows)	111	111
Minimum lead	1 flow (episode 21)	1 flow (episode 21)
Maximum lead	201	191
>= 200 flows early	33.33%	0%
>= 100 flows early	66.67%	66.67%

7. DAY 1 FINAL STATUS

Project status at end of Day 1. Status labels are conservative on purpose.

AREA	STATUS	DETAILS
Dataset audit	COMPLETED	Exploration, label inventory, DDoS-focused inspection
DDoS feature preparation	COMPLETED	Prepared feature set and temporal analysis
Leakage-safe forecasting windows	COMPLETED	V2: 1,911 windows; 0 DDoS in observation
Sequence tensors	COMPLETED	V2 tensors (1365/273/273) x (100, 68); NaN/Inf = 0
Logistic Regression baseline	COMPLETED	TEST F1 0.4091; classical reference
GRU V1	COMPLETED	TEST F1 0.6667; recall 1.00; FAR 0.1235
GRU V1 early warning	COMPLETED	3/3 episodes warned; min lead = 1 flow
GRU V2	COMPLETED	TEST F1 0.6818; recall 1.00; FAR 0.1152
GRU V2 early warning	COMPLETED	3/3 warned; episode 21 still 1-flow lead
Multi-horizon windows	COMPLETED	18,921 windows; four horizons; safety check = 0

AREA	STATUS	DETAILS
Multi-horizon GRU	NOT YET COMPLETED	Prepared for Day 2; no confirmed trained results
Website	COMPLETED	React dashboard in website/; mock data still present
Backend / API	IN PROGRESS	Contract still to be finalised; not a live ML API
GitHub integration	COMPLETED	Pushed on feature/pulkit-integration @ ec43af8

Project status snapshot

V2 GRU Best confirmed sequence model	Windows Multi-horizon data ready	Website UI ready; backend not live	Day 2 Train multi-horizon GRU
--	--	--	---

8. DAY 2 — ADAPTED EXECUTION PLAN

Central objective: Move from single-horizon proof-of-concept forecasting toward validated multi-horizon forecasting and prepare the system for integration.

This plan is based on the actual Day 1 state. It does not blindly repeat Isolation Forest as the next ML step, and it does not ask Pulkit to start a dashboard from zero.

8.A Member allocation

Member	Day 2 tasks	Deliverable
Pranshu Cybersecurity	Document 3-4 concrete attack patterns (DDoS and relevant behaviour); traffic signatures; before/during/after timeline; Detection vs Forecasting; MITRE ATT&CK where useful	Attack-pattern + traffic-signature document
Pulkit Web / integration	Review React dashboard; finalise API/data contract (/predict, /traffic-stream, /alerts, /model-stats); define prediction, alert, traffic-stream, and model-statistics schemas; inventory mock/static values; prepare UI for a future real backend; do not present mock data as ML output	API contract + dashboard integration specification
Pragati Data engineering	Validate multi-horizon windows and leakage prevention; analyse temporal/network features (flow duration, byte rate, protocol, flags); document feature meanings; support sequence preparation; verify preprocessing/scaling consistency	Feature-analysis + validated multi-horizon data preparation
Ankita ML lead	Lead multi-horizon GRU work: tensors, training, 50/100/200/500 evaluation, architecture choices, horizon comparison; coordinate output format with Pulkit. Do not invent results before training.	First validated multi-horizon forecasting model/results
Riddhi Evaluation	Reusable evaluation pipeline comparing LogReg, GRU V1, GRU V2, and multi-horizon GRU; Precision/Recall/F1/ROC-AUC/PR-AUC/FAR plus lead-time metrics; confusion matrices, tables, plots; honest limitations	Reusable evaluation framework + model comparison table
Priyanshi Python / automation	Automate data-loading and repeatable validation; episode-split checks; output schema checks; training/evaluation automation; reproducibility; confirm model outputs can be consumed by a future API	Repeatable automation/validation workflow

8.B Original roadmap vs adapted Day 2

Member	Original Day 2	Adapted Day 2 (based on actual Day 1)
Pranshu	Attack patterns + signatures	Same, now grounded in the DDoS forecasting demo timeline
Pulkit	API routes + mock dashboard	Review existing dashboard; freeze schemas; mark mock vs real
Pragati	Generic feature analysis	Validate multi-horizon windows/tensors and leakage

Member	Original Day 2	Adapted Day 2 (based on actual Day 1)
Ankita	Isolation Forest baseline	Multi-horizon GRU (IF is optional later, not the critical path)
Riddhi	Evaluation skeleton	Full comparison framework across confirmed models + new GRU
Priyanshi	Loader + time split	Reproducible validation/training automation and schema checks

9. Day 2 critical path, timeline, and definition of done

Highest priority: Multi-horizon sequence validation -> Multi-horizon GRU -> Evaluation

Dependencies

Order	Owner	Handoff
1	Pragati	Validated multi-horizon data / tensors; leakage confirmed
2	Ankita	Trained multi-horizon GRU or a documented blocker
3	Riddhi	Horizon metrics + comparison against LogReg / GRU V1 / GRU V2
4	Pulkit	Integration-ready prediction/alert/stats schema using real field names

Pranshu (attack scenarios) and Priyanshi (automation) work in parallel and should not wait on GPU/CPU training to start.

Timeline

Block	Focus
Morning	Data/tensor validation; feature analysis; attack-pattern work; API contract review; evaluation framework preparation
Midday	Multi-horizon GRU training; evaluation pipeline; dashboard preparation; cybersecurity scenario documentation; automation
Afternoon	Model evaluation; horizon comparison; dashboard forecasting visualisation planning; results consolidation
End of day	15-minute team sync; consolidate results; identify blockers; Git checkpoint; prepare Day 3 inputs

Day 2 definition of done

- ☐ Multi-horizon windows validated
- ☐ No observation leakage confirmed
- ☐ Multi-horizon sequence tensors validated
- ☐ Multi-horizon GRU trained OR blocker documented
- ☐ 50-flow results available OR blocker documented
- ☐ 100-flow results available OR blocker documented
- ☐ 200-flow results available OR blocker documented
- ☐ 500-flow results available OR blocker documented
- ☐ Model comparison framework ready
- ☐ Attack patterns documented
- ☐ Feature analysis documented
- ☐ API contract finalised
- ☐ Dashboard mock/real-data boundaries documented
- ☐ Automation checks completed
- ☐ End-of-day Git checkpoint completed

10. Risks and honest limitations

#	Risk / limitation	Why it matters on Day 2
1	Only 3 held-out TEST episodes (19-21) are used for episode-level early-warning analysis	Do not generalise lead-time statistics as if they were a large operational trial

#	Risk / limitation	Why it matters on Day 2
2	100% warning rate does not mean consistently useful early warning	Always report lead-time distribution, not only warning rate
3	GRU V2 (and V1) had one TEST episode with only 1-flow lead	A late warning is operationally close to detection, not forecasting
4	False alarms remain important (GRU V2 TEST FAR = 0.1152)	Precision and FAR must sit beside recall in any demo narrative
5	Multi-horizon GRU results are not available at the end of Day 1	Day 2 critical path; do not fill slides with invented horizon metrics
6	Frontend dashboard is not yet proven connected to a real ML backend	Label mock widgets clearly; freeze API schemas before wiring
7	Raw CICIDS2017 and large model/tensor artifacts are excluded from GitHub	Teammates need a local data/model sharing plan, not only git pull
8	The system must demonstrate forecasting, not only static attack classification	Keep observation-window safety and lead time as first-class demo metrics

What this backup does not cover

This document stops at Day 1 results and the Day 2 plan. It does not schedule Days 3-13, and it does not treat feature/pulkit-integration as already merged to main.

Closing note for the team

Day 1 produced a leakage-aware DDoS forecasting pipeline, two trained GRU models with measured early-warning behaviour, a completed multi-horizon window dataset, a React dashboard, and a pushed feature branch. Day 2 is not another kickoff. It is the first day that must turn multi-horizon data into a validated model, an honest comparison, and an integration contract — without overclaiming warning quality or dashboard live-ness.

End of internal backup report. SIH26153 | 1 September 2026 | Commit ec43af8